

MHF 4U1- Unit 5 Test - Trigonometric Functions

Name _____

Date _____

Total

24.5

/30

KNOWLEDGE

1. Solve for x in the interval $[0, 2\pi]$. Give the **exact solution(s)** in radian measure, if possible. Otherwise, round to the nearest hundredth of a radian.

a) $\cos x = \tan x \cos x$

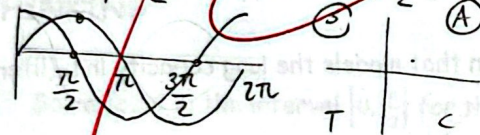
$$\cos x = \frac{\sin x}{\cos x} (\cos x)$$

$$\frac{\cos x^2 - \sin x \cos x}{\cos x} = 0$$

$$\cos x (1 - \sin x) = 0$$

$$\downarrow \cos x = 0$$

$$x = \frac{\pi}{2}, \frac{3\pi}{2}$$



$$\left| \frac{\pi}{2}, \frac{3\pi}{2} \right|$$

$$- \sin x + 1 = 0$$

$$\sin x = 1$$

$$x = \frac{\pi}{2}$$

$$\text{A}$$

$$\cos x - \tan x \cos x = 0$$

$$\cos x (1 - \tan x) = 0$$

$$\cos x = 0$$

$$\text{for } \tan = 1$$

$$\text{and } \cos x$$

b) $6 \sec^2 x = 11 \sec x - 4$

let $a = \sec x$

$$6a^2 - 11a + 4 = 0$$

$$(3a - 4)(2a - 1) = 0$$

$$\sec x = \frac{4}{3}$$

$$\cos x = \frac{3}{4}$$

$$\cos^{-1}\left(\frac{3}{4}\right) = 0.722$$

$$\text{A}$$

$$\text{T}$$

$$\text{C}$$

$$Q_1: 0.722$$

$$Q_4: 2\pi - 0.722$$

$$= 5.56$$

$$d) \sqrt{5} \tan\left(\frac{x}{2}\right) - 4 = 0$$

$$\tan\left(\frac{x}{2}\right) = \frac{4}{\sqrt{5}}$$

$$\tan^{-1}\left(\frac{4}{\sqrt{5}}\right) = 1.061$$

$$Q_1: \frac{x}{2} = 1.061$$

$$x = 2.12$$

$$\text{not } \div 2$$

$$Q_2: \frac{x}{2} = \pi + 1.061$$

$$x = 2.10$$

$$\text{A}$$

$$\text{T}$$

$$\text{C}$$

$$\left| 0.53, 2.10 \right|$$

2

(3)

(3)

$\cos^{-1}(2) = \text{no solution}$
bc $\cos > 2$ is invalid

$$\left| 0.72, 5.56 \right|$$

S₂

A₁

T₂

C₄

don't be dumb

x2

not

div 2

tr

(3)

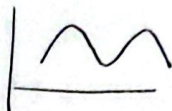
(3)

(3)

2.5

10.5

APPLICATION



1. A Ferris wheel ride has a height, h , in metres, that varies over time, t , in seconds from the start of the ride. The ride can be modelled by the equation $h(t) = 5\sin\left[\frac{\pi}{6}(t-4)\right] + 6$. List the four consecutive times in seconds when the Ferris wheel hits height of 10 m.

$$10 = 5\sin\left[\frac{\pi}{6}(t-4)\right] + 6$$

$$\frac{10-6}{5} = \sin\left[\frac{\pi}{6}(t-4)\right]$$

$$\frac{4}{5} = \sin\left[\frac{\pi}{6}(t-4)\right]$$

$$\text{let } a \text{ rep. } \left[\frac{\pi}{6}(t-4)\right]$$

$$\frac{4}{5} = \sin a$$

$$\sin^{-1}\left(\frac{4}{5}\right) = 0.927$$

$$\begin{array}{c|c} \textcircled{S} & \textcircled{A} \\ \hline t & c \end{array}$$

$$Q_1: \frac{\pi}{6}(t-4) = 0.927$$

$$t = 5.7705$$

$$Q_2: \frac{\pi}{6}(t-4) = \pi - 0.927$$

$$t = 8.2295$$

$$\text{period: } \frac{2\pi}{\frac{\pi}{6}} = 12$$

$$\text{third time: } 5.7705 + 12 = 17.7705$$

$$\text{fourth time: } 8.2295 + 12 = 20.2295$$

$$5.7705, 8.2295, 17.7705, 20.2295$$

2. When seated, an average adult breathes in and out 15 times in one minute. The average minimum amount of air in the lungs is 0.08 L, and the average maximum amount of air in the lungs is 0.88 L. Suppose that lungs have a minimum amount of air at $t = 0$, where t is the time in seconds.

- a) Graph two complete cycles of the function. Write a sine function that models the lung capacity in L (liters) over time (seconds). Be sure to label the axes.

$$\frac{60s}{15s} = 4s$$

1 cycle

$$k = \frac{2\pi}{4} = \frac{\pi}{2}$$

$$\text{amp: } \frac{0.88 - 0.08}{2}$$

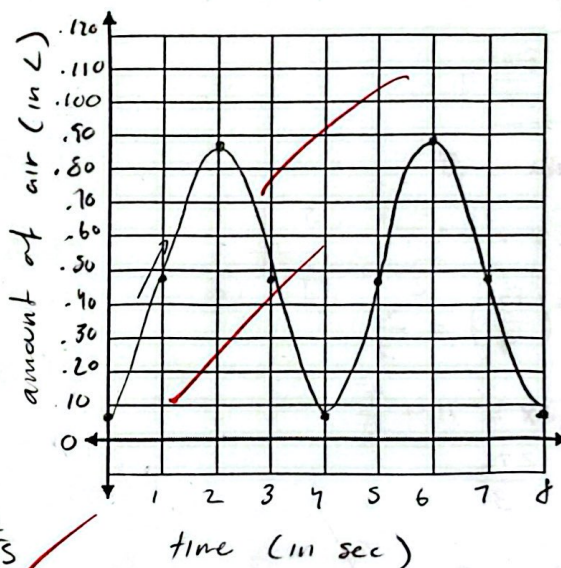
$$= \frac{2}{5}$$

$$\text{axis: } \frac{0.88 + 0.08}{2}$$

$$= \frac{12}{25}$$

$$d = +1$$

$$\text{Equation: } f = \frac{2}{5}\sin\left[\frac{\pi}{2}(t-1)\right] + \frac{12}{25}$$



- b) At what time(s) does the instantaneous rate of change reach a maximum? Calculate the maximum instantaneous rate of change.

IROC reaches max value @ $t = 1, 3, 5, 7$ (min values)

$$\text{IROC @ } t=1: \frac{0.480628 - \frac{12}{25}}{0.001 - 1}$$

$$\text{IROC @ } t=3: \frac{0.479371 - \frac{12}{25}}{3.0001 - 1}$$

$$\text{IROC @ } t=5: \frac{0.480628 - \frac{12}{25}}{5.001 - 1} = 0.628 \text{ L/s}$$

$$\text{IROC @ } t=7: \frac{0.479371 - \frac{12}{25}}{7.0001 - 1} = -0.629 \text{ L/s}$$

$$\text{IROC @ } t=1: \frac{0.480628 - \frac{12}{25}}{0.001 - 1} = 0.628 \text{ L/s}$$

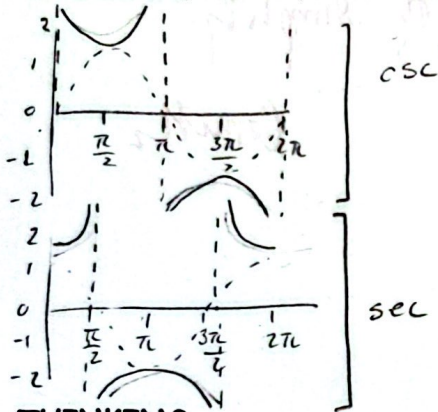
$$\text{IROC @ } t=3: \frac{0.479371 - \frac{12}{25}}{3.0001 - 1} = -0.629 \text{ L/s}$$

COMMUNICATION

1. Explain the difference between $\sec \frac{1}{\sqrt{2}}$ and $\cos^{-1}(\frac{1}{\sqrt{2}})$.

$\sec \frac{1}{\sqrt{2}}$ is the reciprocal of the $\cos$ function, and is using the given angle of $\frac{1}{\sqrt{2}}$ to find a ratio. It is equivalent to $\frac{1}{\cos x}$ so the angle would be flipped to $\frac{\sqrt{2}}{1}$ and used to find the ratio. However, $\cos^{-1}(\frac{1}{\sqrt{2}})$ is trying to find an angle with a given ratio by doing the inverse.

2. Describe, in detail, two similarities and one difference between the graphs of cosecant and secant.



THINKING

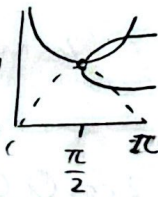
one difference is that due to the fact reciprocal graphs have asymptotes @ the primary graph's 0s, $\csc$ and $\sec$ have different asymptotes. $\csc$ has asymptotes @ $x = 0, \pi, 2\pi$ and $\sec$ has @ $x = \frac{\pi}{2}, \frac{3\pi}{2}$. 2 similarities the reciprocal graphs have is they have the opposite slope to their primary graph. For example, from $[0, \frac{\pi}{2}]$ is (+) slope on the

1. Solve for θ in the interval $[0, \frac{\pi}{2}]$ for the trigonometric equation, $\tan\theta(1 + \cos 2\theta) = 2\sin^2 2\theta$. Show all steps.

ON BACK

do not compare to primary graph, look solely

$\sin$ graph but its (-) slope on the $\csc$ graph. Both $\csc x$ and $\sec x$ graphs follow this. Another similarity is that the reciprocal graphs meet their primary graphs at a max/min point but it becomes the opposite local max/min. For example, Both reciprocal graphs follow this



local min for $\csc x$
local max for $\sin$

blurb $\sec x$ and $\csc x$

$$\tan \theta (1 + \cos 2\theta) = 2 \sin^2 2\theta \quad \left[0, \frac{\pi}{2} \right]$$

$$\tan \theta + \tan \theta \cos 2\theta = 2 \sin^2 2\theta$$

$$\frac{\sin \theta}{\cos \theta} + \frac{\sin \theta \cos 2\theta}{\cos \theta} = 2 \sin^2 2\theta$$

$$\frac{\sin \theta + \sin \theta (1 - 2 \sin^2 \theta)}{\cos \theta} = 2 \sin^2 2\theta$$

not able
to simplify
correctly

$$\frac{\sin \theta + \sin \theta - 2 \sin^3 \theta}{\cos \theta} = 2 (2 \sin \theta \cos \theta)^2$$

$$\sin \theta + \sin \theta - 2 \sin^3 \theta = 2 \cos \theta (2 \sin \theta \cos \theta)^2$$

$$2 \sin \theta - 2 \sin^3 \theta = 2 \cos \theta (4 \sin^2 \theta \cos^2 \theta)$$

$$2 \sin \theta - 2 \sin^3 \theta = 8 \sin^2 \theta \cos^3 \theta$$

$$0 = 8 \sin^2 \theta \cos^3 \theta + 2 \sin^3 \theta - 2 \sin \theta$$

$$= \sin \theta (8 \sin \theta \cos^3 \theta + \sin^2 \theta - 2)$$

$$\tan \theta (1 + \cos 2\theta) = 2 \sin^2 2\theta \quad \left[0, \frac{\pi}{2} \right]$$

$$\frac{\sin \theta}{\cos \theta} + \frac{\sin \theta \cos 2\theta}{\cos \theta} = 2 (2 \sin \theta \cos \theta)^2$$

$$(2 \cos^2 \theta - 1)(2 \cos^2 \theta - 1)$$

$$\frac{\sin \theta + \sin \theta \cos 2\theta}{\cos \theta} = 2 (2 \sin \theta \cos \theta)^2$$

$$\sin \theta + \sin \theta \cos 2\theta = 2 \cos \theta (2 \sin \theta \cos \theta)^2$$

$$\sin \theta + \sin \theta (\cos^2 \theta - \sin^2 \theta) = 4 \cos^3 \theta \sin^2 \theta$$

$$\sin \theta + \sin^3 \theta + \sin \theta \cos^2 \theta - 4 \cos^3 \theta \sin^2 \theta = 0$$

$$\sin (\sin^2 \theta + \cos \theta - 4 \cos^3 \theta \sin \theta) = 0$$

just anton to
send when he figures it out